

Enhanced Detection of Age-Related and Cognitive Declines Using Automated Hippocampal-to-Ventricle Ratio in Alzheimer's Patients

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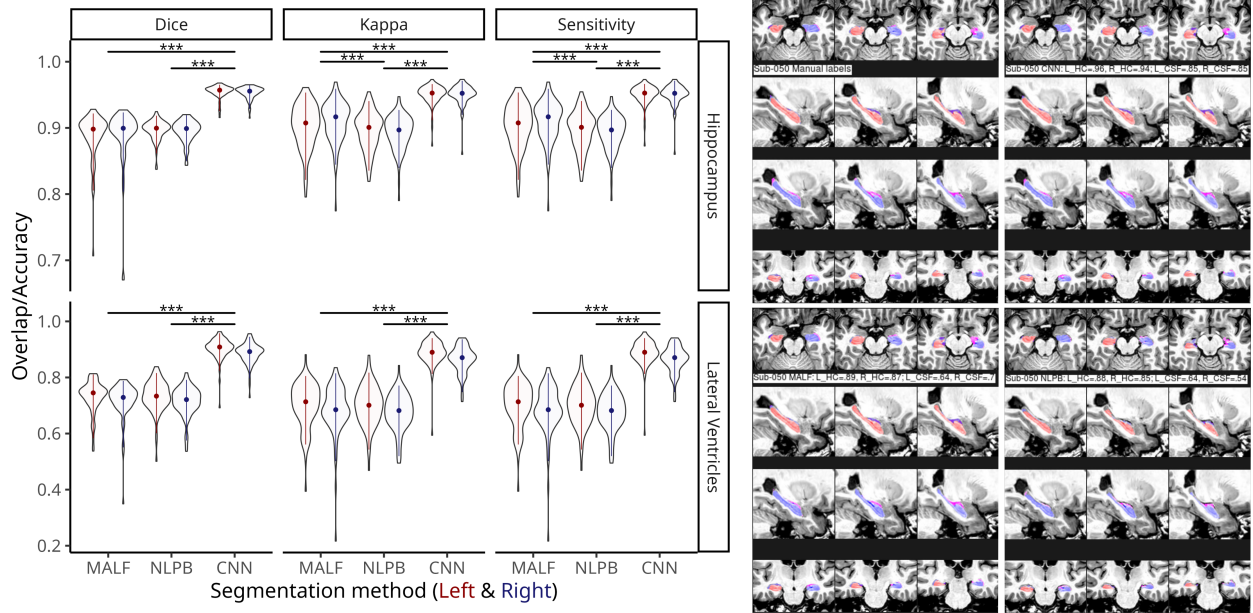


FIGURE 1 – Automatic segmentation performance.

Left: violin plots of the distributions of the overlap and accuracy metrics (Dice, Kappa, and Sensitivity) between manual and computed segmentations of the hippocampus and ventricles from the left (red) and right (blue) hemispheres using MALF, NLPB and CNN. The median and standard deviation are represented inside the violin plots. Comparison lines above the violin plots show statistical pair-wise difference between the segmentation methods ($*** = p < 0.001$). Right: Mosaic images of the manual and computed segmentations for an example subject.

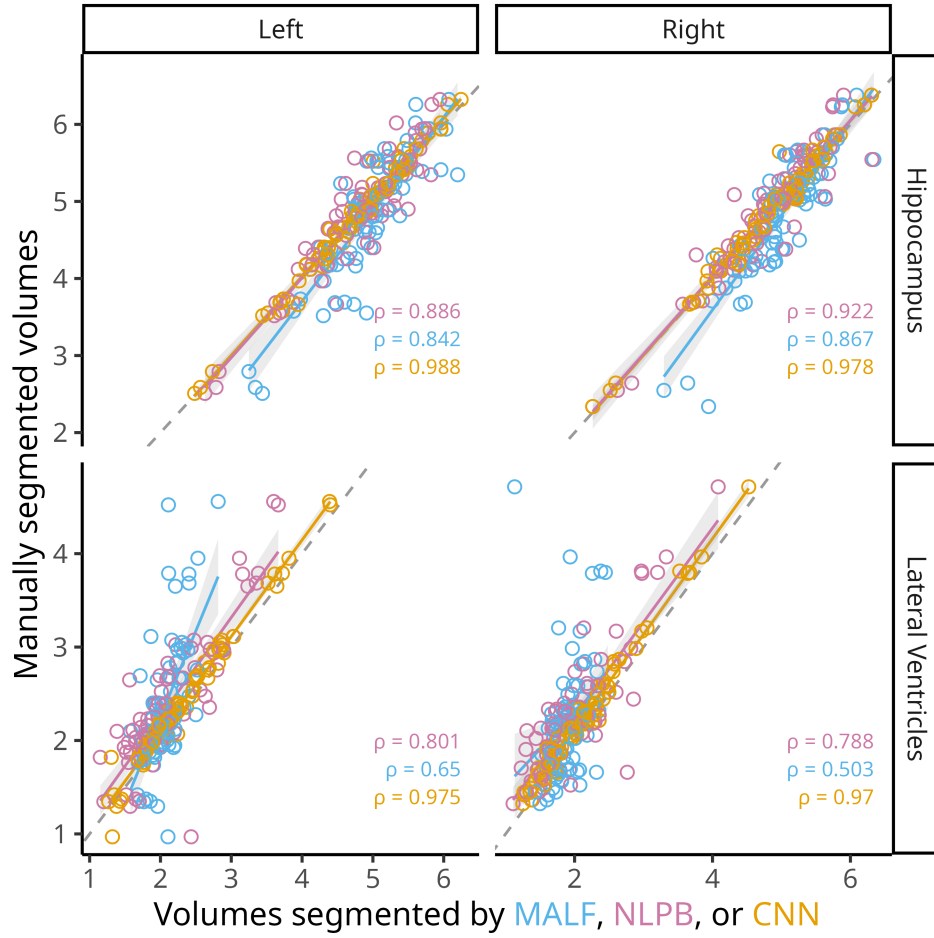


FIGURE 2 – Relationship between manual and computed hippocampal (HCvol) and ventricular (CSFvol) volumes.

Scatter plots of the volumes (measured in cc) of the left and right hippocampus and ventricular space from the manual segmentations against those obtained from MALF (blue), NLPB (green), and CNN (orange). The Spearman's rank correlation coefficients (ρ) and their significance are added for each automatic method.

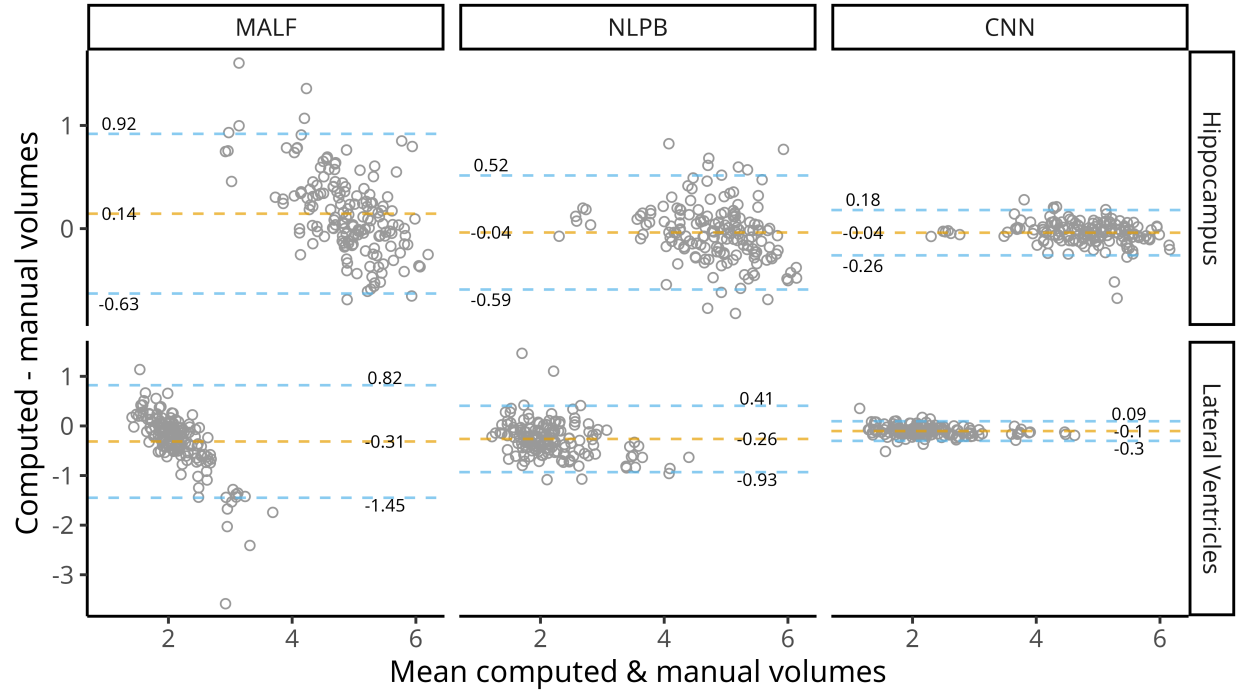


FIGURE 3 – Bland-Altman plots.

Bland-Altman plots for the volumetric comparison between mean HCvol and CSFvol (in cc) resulting from the automatic segmentation methods and manual labels. The dashed lines represent the mean difference (orange) and the upper and lower limits of agreement (blue).

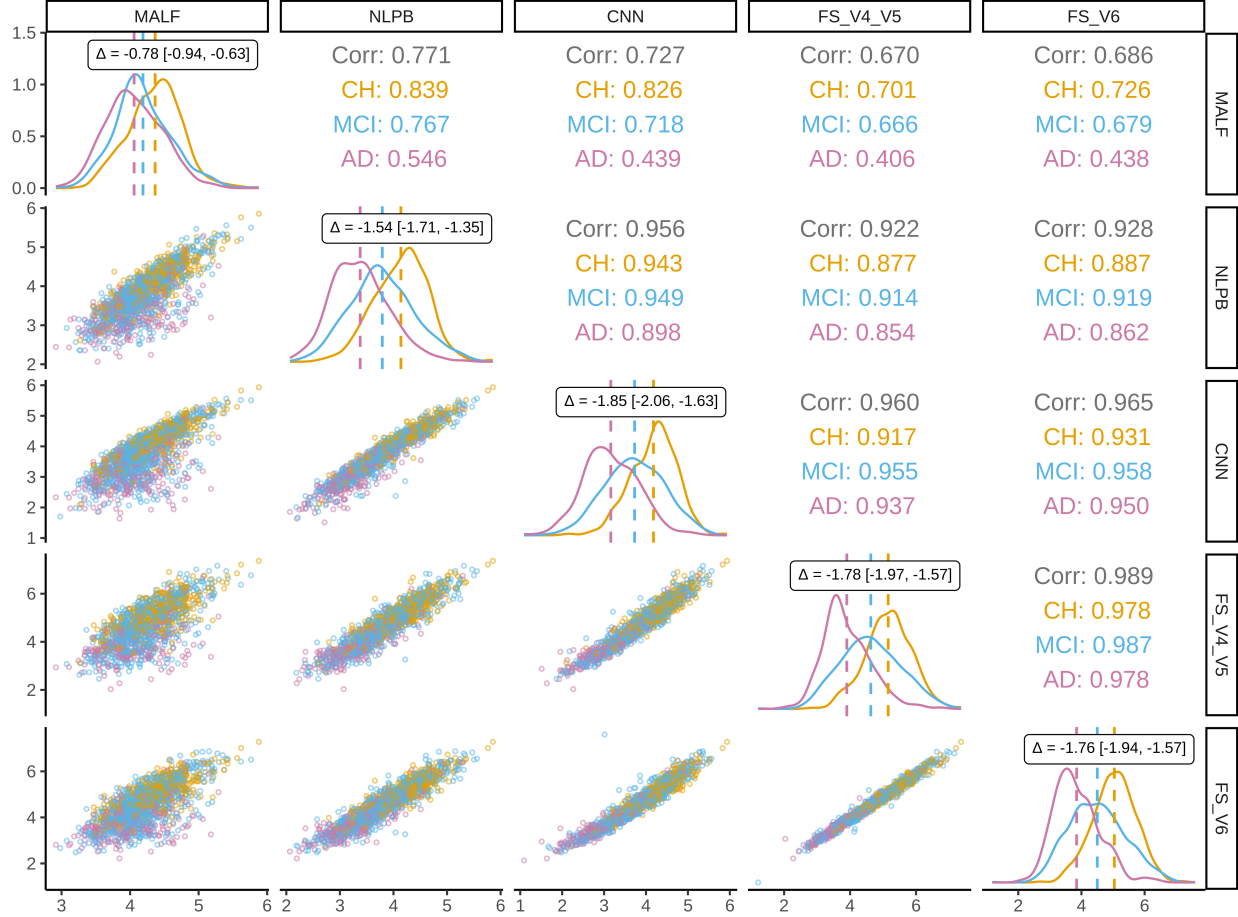


FIGURE 4 – Similarity of segmentation and clinical effect sizes of HCvol.

The lower triangle of the matrix plot shows scatter plots of the HCvol values across the different automatic segmentation methods stratified by the clinical group (CH: orange, MCI: blue, and AD: green). The upper triangle of the matrix plots lists the correlation coefficients between the automatic segmentation methods using Spearman's rank correlation. The correlation coefficients (ρ) include the values for the whole sample in gray and by clinical group. The diagonal of the matrix plot shows the density plots of the HCvol by clinical group. Dashed lines represent the mean of each cohort. Inside each density plot, the inlaid box shows calculated effect sizes between the CH and AD groups using Glass Delta and their robust 95% confidence intervals.

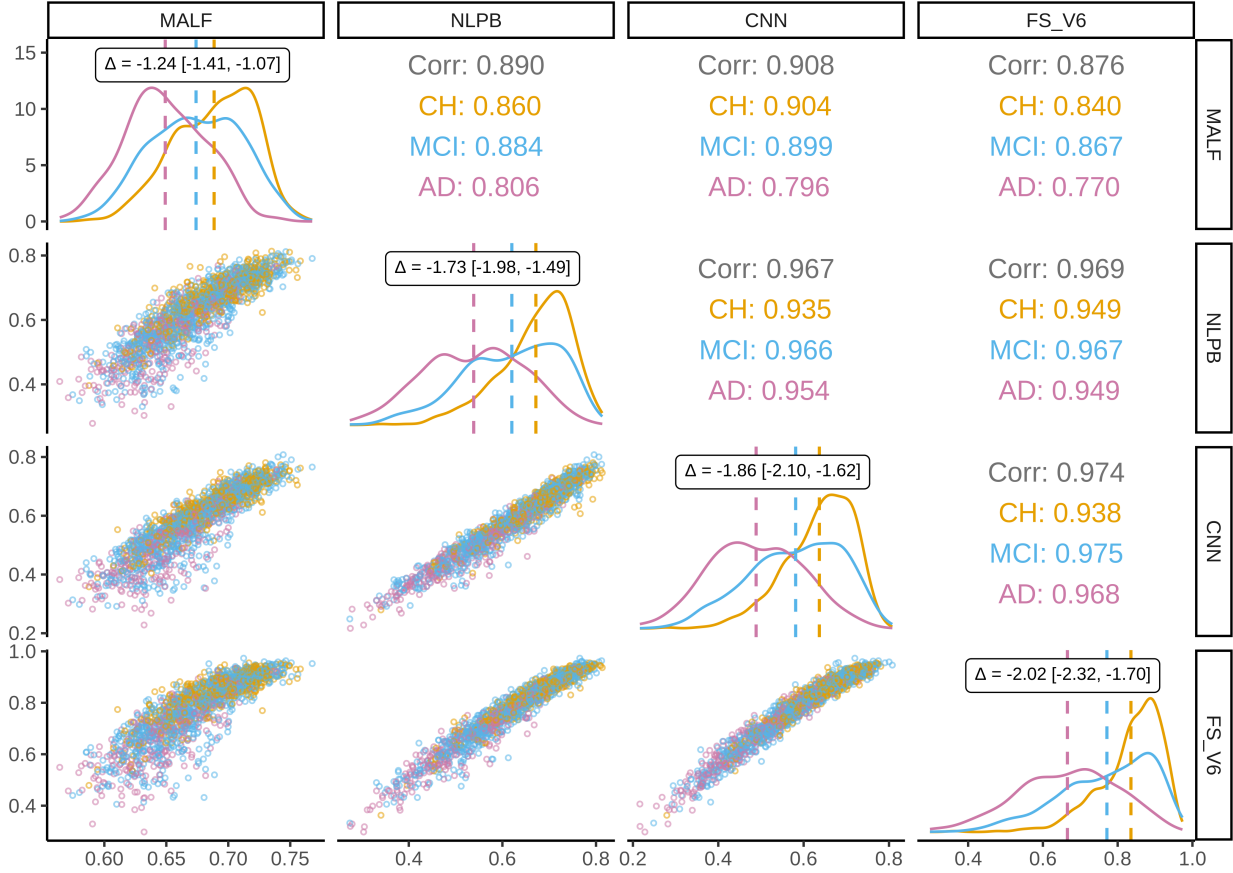


FIGURE 5 – Similarity of values of HVR and their derived clinical effect sizes.

The lower triangle of the matrix plot shows scattered plots of the HVR values across the different automatic segmentation methods stratified by the clinical group (CH: orange, MCI: blue, and AD: green). The upper triangle of the matrix plot lists the correlation coefficients (ρ) between the automatic segmentation methods using Spearman's rank correlation. The correlation coefficients include the values for the whole sample in gray and by clinical group. The diagonal of the matrix plot shows the density plots of the HVR by clinical group. Dashed lines represent the mean of each cohort. Inside each density plot, the inlaid box shows calculated effect sizes between the CH and AD groups using Glass Delta and their robust 95% confidence intervals.

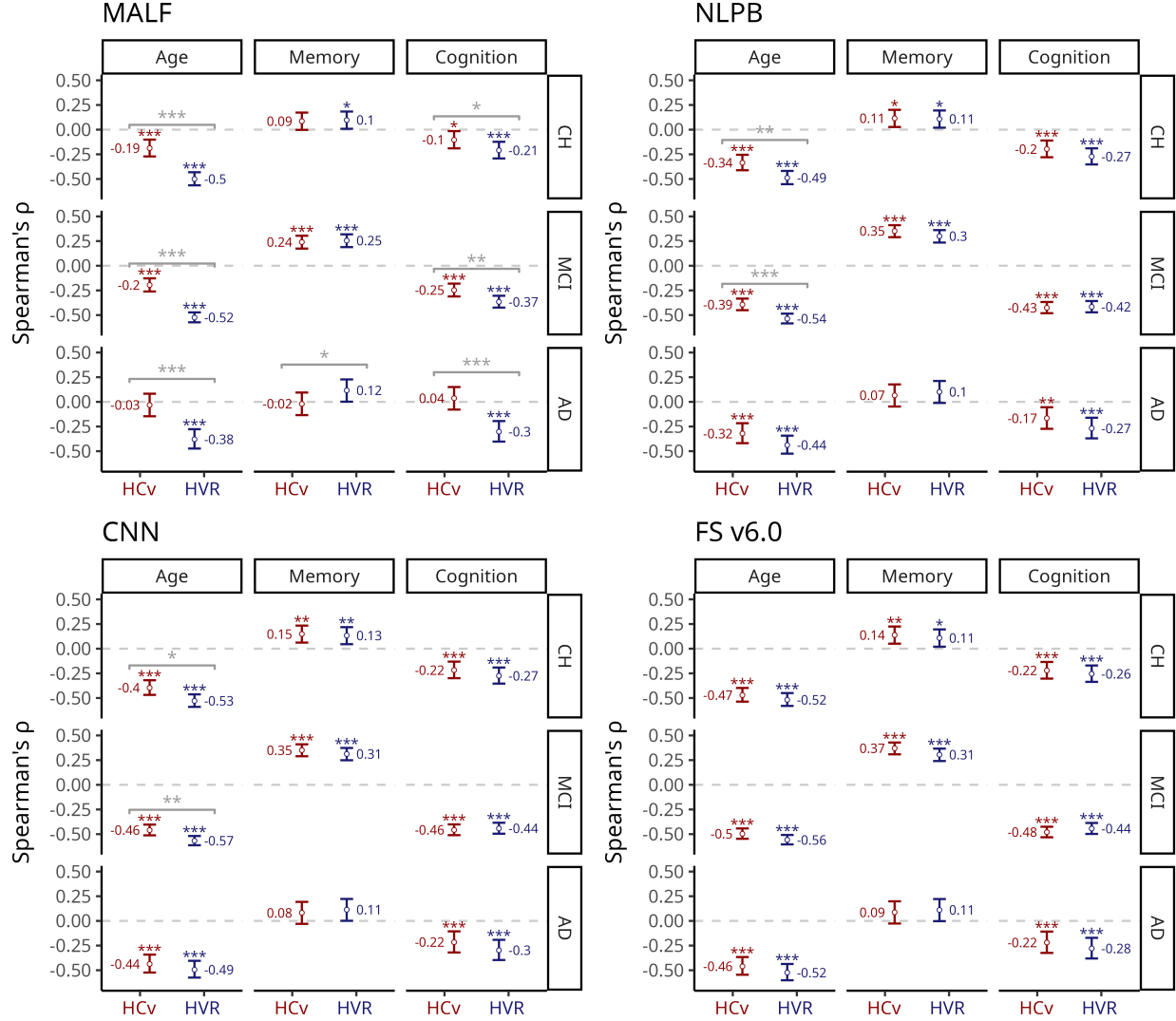


FIGURE 6 – Associations between hippocampal measures and age, memory, and global cognition.

Summary of the correlation coefficients measured by Spearman's rank correlation coefficients (ρ), between HCvol (HCv in the legend), HVR, and age, memory (measured by RAVLT learning subscore), and global cognition (measured by ADAS13). Colored asterisks mark the significance of the correlation (* = $p < 0.05$; ** = $p < 0.01$, *** = $p < 0.001$). Gray asterisks mark the significance comparing HCvol and HVR. All p-values are corrected for multiple comparisons.